

HyMeKo — Capability Evidence Ledger

A calibrated inventory — claims tied to auditable artifacts, not ambition

Compiled 2026-06-20 | Companion: *The HyMeKo System-Engineering View* (manifesto)

The standard this document holds itself to

Each row separates three things on purpose: **the specific claim**, **the artifact that checks it**, and — where it matters — **what the artifact does not establish**. Any line can be audited by opening the named file.

Two honesty rules. (1) A claim is only as strong as what is *mechanically* checked: “a test passes” means its own assertions hold, and does not silently extend to properties the test never examines. Where a test proves structure but not visual fidelity or numerical parity, the row or the section note says so. (2) **Provenance is marked.** Results re-verified this session are tagged (*measured*); results resting on a prior dated report are attributed to it rather than presented as freshly confirmed.

Status legend. **PROVEN** — a passing test or committed artifact asserts the specific claim. **SHIPPED** — works in a demo/report but is not pinned by a regression test, so it can regress silently. **RESEARCH** — an open investigation, reported as positive, partial, null, or falsified — whichever it is.

A status is never upgraded without an anchor; a row with a stale anchor is a defect, not a convenience.

1. Hypergraph core — the single source of record

Capability	What the evidence actually establishes	Status
Signed-incidence hypergraph; GGK 4-tuple $K = (B, G, \mu, r)$ with signed-incidence tensor	Core types + the K1–K4 axioms they satisfy are specified in docs/spec/g_sphf_axioms.tex (cited in published work) and implemented in locked <code>hymeko_core</code> . The axioms are a design contract, not a machine-checked proof.	PROVEN (impl)
Canonical hashing of descriptions	<code>canonical_hash.rs</code> : hash is invariant under semantics-preserving rewrites and changes under semantic ones; parity asserted across suites, so an accidental semantic change surfaces as a failing test.	PROVEN
Lower $\rightarrow$ resolve $\rightarrow$ intern compile pipeline (<code>ModuleStore</code>)	Production compile path exercised by the <code>hymeko_core</code> suite (≈ 133 tests per the 2026-06-19 report). Not independently re-run this session.	PROVEN (per suite)

2. The HyMeKo language

Capability	What the evidence actually establishes	Status
LALR(1) grammar + parser (locked, generated tables)	Round-trips the fixture corpus including alias resolution (<code>using_alias.rs</code>). Tables are generated, never hand-edited.	PROVEN

Capability		What the evidence actually establishes	Status
Profiles, <code><isa></code> inheritance, tag annotations, tensor/edge values		<code>data/minimal_examples/**</code> + core import tests parse and lower each construct — proves the language accepts and lowers them, not that every semantic edge case is covered.	PROVEN
Cross-file (<code>@"file.hymeko"</code>) meta vocabulary	imports sharing	<code>test_import/</code> + <code>import_examples/</code> compile a file against vocabulary declared in another. Establishes kind-sharing across files.	PROVEN
Cross-profile references (<code><desc>.<content></code> as <code>a</code> ; (+ <code>a.decl</code>))	instance (using	Core test <code>check_xprofile_instance_ref</code> : the referenced <code>dist decl</code> 's hash is <i>unchanged</i> vs. the inline form — concrete evidence semantics were preserved, not merely that it compiled (report 2026-06-19).	PROVEN
Import-aware bundle reading on the Python side (<code>read_bundle</code>)		Mirrors the compiler's import resolution so Python tooling sees the same merged <code>decl</code> set. Stated limit: importer must also import the meta vocab the shared profile uses (no transitive indexing yet).	PROVEN (w/ limit)

3. Robotics — kinematic extraction + multi-format emission

Capability		What the evidence actually establishes	Status
Kinematic-model extraction		Recovers links, typed joints, axis vectors, limits, geometry, origins. <code>test_imported_real</code> asserts counts within a documented filter tolerance: ≥ 8 links / ≥ 6 joints (WAM), ≥ 50 / ≥ 50 (DRC-Hubo).	PROVEN
Emitters: URDF, SDF, MJCF, DOT, Mermaid, SysML (one IR $\rightarrow$ six views)		Six emitters produce non-empty output carrying the correct robot header on real fixtures; <code>test_mdsc_reuse</code> additionally asserts URDF contains real <code><joint></code> elements with recovered unit axis vectors (<i>measured</i>).	PROVEN (structure)
Lean4 proof-obligation emission		<code>emit_lean4.rs</code> + tests emit non-empty Lean4 from the IR. Proves emission, not that the obligations discharge.	PROVEN (emit)
HyMeKo $\rightarrow$ HyMeKo canonical round-trip		<code>emit_hymeko.rs</code> + <code>test_bridge.rs</code> re-emit and check back: the information-loss guard for any view generator.	PROVEN
Emission on real published robots (Barrett WAM 7-DOF; DRC-Hubo 52-link humanoid) via import		Translated URDFs compile + emit through the full pipeline — the “not only synthetic morphologies” evidence.	PROVEN
MDSD reuse for a full scenario	single-source	(<i>measured 2026-06-20</i>) imported form $\Rightarrow$ 5 links + 4 revolute joints with unit axes, emits <code><joint>s</code> ; the bare baseline's generic <code>joint</code> emits none; 126 vs 189 lines (-33%).	PROVEN

Capability	What the evidence actually establishes	Status
Gazebo launch-bundle generation	<code>test_gazebo_sim_launch</code> builds URDF + world + launch; committed bundle at <code>generated/gazebo_launch/moveo/</code> .	PROVEN
Robot generators (anthropomorphic arms; chain/tree/quadruped/humanoid scaling fixtures)	Generation tests build valid descriptions across morphologies and sizes.	PROVEN

Scope — what §3 does not claim. Emission is structural and kinematic, not visual: the importer strips mesh references and substitutes placeholder box geometry (documented in `test_imported_real.rs`), so a round-tripped robot will not visually reproduce the original; mesh-bearing emission is a separate, unbuilt line. Lean4 output is emitted, not proven to discharge.

4. Query, rewrite, analysis

Capability	What the evidence actually establishes	Status
Predicate query engine over the IR (<code>node/edge/inherits/and</code>)	Drives all of §3’s extraction; the full <code>hymeko_query</code> integration target runs 214 pass , 1 ignored (<i>measured 2026-06-20</i>).	PROVEN
Structural entropy + cluster-split rewrite (emits a proposed <code>.hymeko</code>)	<code>test_entropy.rs</code> , <code>test_split.rs</code> compute entropy and a split that recompiles.	PROVEN
Cycle/Berge enumeration, Friedler pruning, branch-and-bound scoring	<code>hymeko_graph</code> + <code>test_fixture_berge.rs</code> enumerate against fixtures with known answers.	PROVEN

5. P-graph (process-network) layer

Capability	What the evidence actually establishes	Status
P-graph schema + A1–A5 axioms; MSG / SSG / ABB	<code>axiom_witness.rs</code> , <code>byproduct_filter_phase11.rs</code> exhibit axiom witnesses and the algorithmic constructions.	PROVEN
Book validation against Friedler / Pimentel worked examples	Suite reproduces the textbook Examples (3.2, 6.1, ...) from <code>data/pgraph/Chapter*/</code> — matches published answers, the strongest correctness evidence available here.	PROVEN
Audit protocols $\leftrightarrow$ reachability rules (leakage audit + P-graph axioms)	Plan + argument + tests exist (2026-06-14); a candidate unification, not yet an external result.	RESEARCH

6. ROS2 / simulation demos

Capability	What the evidence actually establishes	Status
Multi-context (grasping / maintenance / safety) context-evaluation node	Runs in <code>hymeko_ros2_demo</code> with <code>test_evaluate_context.py</code> , <code>test_topic_binding.py</code> .	SHIPPED
UR5e Gazebo + MoveIt2 stack; dual-robot handover	Launch files + sim assets present and demoed.	SHIPPED
MuJoCo RL grasp/reach env emitted from <code>.hymeko</code>	Built and run in the 2026-06-19 reports.	SHIPPED

*Scope. These are **SHIPPED**, not **PROVEN**: they work in the demo environment and have Python unit tests around pieces, but no single regression test pins the end-to-end ROS2/Gazebo path, so it can regress silently on an environment change.*

7. Reinforcement learning — descriptions → MDP

Capability	What the evidence actually establishes	Status
Declarative observation / reward specs in HyMeKo	The MDP's observation and reward are authored as <code>.hymeko</code> , not Python; reports 2026-06-19-hymeko-rl-declarative-*	PROVEN
<code>AgentSpec.from_hymeko</code> (obs + action bounds + reward + vertex count → one MDP)	Composes a complete MDP from robot + task descriptions; parity test vs. the hand-built env passes (97 <code>hymeko_rl</code> tests green, per report 2026-06-19).	PROVEN (per suite)
Shared reward / observation models reused across agent descriptions	<code>arm_reach_reward.hymeko</code> imported by multiple tasks; reward <code>dist</code> decl hash unchanged vs. the duplicated form (semantics preserved).	PROVEN
PPO fine-tuning on the BC-pretrained policy	Negative-to-open: PPO currently <i>degrades</i> the behaviour-cloned policy; leading hypothesis is a truncation-bootstrap bug, not yet isolated. Paused.	RESEARCH (paused)

8. Tooling surface

Capability	What the evidence actually establishes	Status
CLI (<code>validate</code> , <code>compile</code> , <code>transform</code> , <code>query</code> , <code>entropy</code> , <code>rewrite</code>)	<code>hymeko_cli</code> ; <code>validate</code> on a fresh scenario returned OK (<i>measured 2026-06-20</i>).	PROVEN
In-browser WASM editor with relationship-enriched snapshots	Built via <code>wasm-pack</code> (§1-approved 2026-06-12); runs in <code>docs/editor/</code> . No automated browser regression.	SHIPPED
IPC daemon + client; MCP server	<code>hymeko_daemon/hymeko_client</code> (API frozen), <code>hymeko_mcp</code> present.	SHIPPED

Capability	What the evidence actually establishes	Status
Benchmark corpora + size sweeps	hymeko_bench/corpora/ + artifact generator.	SHIPPED

9. Research lines — reported as found

Capability	What the evidence actually establishes	Status
Signed-link prediction; leakage audit (label-shuffle, Gömb-strict)	Audit harness + vectorisation ($\sim 6\times$) + figure done; 5-seed grid running. The contribution is the <i>audit protocol</i> ; results in progress.	RESEARCH
HSiKAN geometric / triad attention; rotor tuning	A/B head built; closing a ~ 0.02 – 0.03 AUROC gap to SiGAT is the open question, attributed (untested) to attention.	RESEARCH
HyMeYOLO / Soma-vision detection	Mostly negative: Soma-vision falsified for vision in the backlog; held-out VOC mAP ~ 0.015 . Reported as a negative result, not a capability.	RESEARCH (negative)
Cayley/Clifford rotor embedding (inductive, leakage-free)	Positive, honest framing: Pareto-efficient on signed-link — near-best AUROC at $\sim 270\times$ fewer parameters, leakage-free under label-shuffle, 5 datasets $\times$ 5 seeds. Beats DADSGNN/SGCN; trails SiGAT by ~ 0.02 at that size — that trade <i>is</i> the result. 9 unit tests.	RESEARCH (validated)
S^1 (cos,sin) rotor encoding of RL joint angles	Falsified, narrowly: one ablation arm. Revolute joints are range-limited (± 2.5 rad), never reach the $\pm\pi$ wrap, so the periodicity buys nothing (smoke: 0.329 vs 0.240 m). Does <i>not</i> bear on the primitive above — different mechanism, different domain. Reverted.	FALSIFIED (scoped)

Reading §9 fairly. The two rotor rows are deliberately distinct: the embedding primitive is a validated parameter-efficiency result; the RL joint-angle encoding is a single falsified ablation. Neither implies the other. The vision line is reported as negative on purpose — a ledger that hid its null results would not be worth reading.

This ledger earns its keep only by staying true: when a capability changes status, the row and its anchor change together. “SHIPPED, not regression-pinned” is preferred over rounding up to PROVEN.